

Multiple Choice (10 marks)

1. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
2. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) $1.2 \times 10^9 \text{ m/s}$
 - (b) $3.0 \times 10^8 \text{ m/s}$
 - (c) $1.5 \times 10^8 \text{ m/s}$
 - (d) $1.0 \times 10^8 \text{ m/s}$
3. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
 - (a) $mc/2$
 - (b) $mc/(2^{1/2})$
 - (c) mc
 - (d) $(3^{1/2})mc$
4. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
 - (a) 44 Hz
 - (b) 196 Hz
 - (c) 262 Hz
 - (d) 393 Hz
5. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
 - (a) a dye laser
 - (b) a helium-neon laser
 - (c) an excimer laser
 - (d) a ruby laser

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of microwaves.
7. True or False: The electric displacement current through a surface S is proportional to the rate of change of the magnetic flux through S .
8. True or False: In the observer's reference frame, the meter stick takes 2.5 ns to pass the observer when moving at a speed of $0.8c$.
9. True or False: The electric displacement current through a surface S is proportional to the time integral of the magnetic flux through S .
10. True or False: An observer must move at $0.60c$ parallel to a 1.00 m rod to measure its length as 0.80 m due to relativistic length contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A gas cell with an optical path length of 10 cm in a Michelson interferometer shows 100 fringes as it is evacuated. Calculate the original gas's index of refraction and discuss the implications of your findings.
12. Discuss the role of dopants in germanium semiconductors and analyze which types of ions are suitable for creating n-type semiconductors, providing justification for your choices.

End of Paper